

Textbook Link

<http://webeducation.com/wp-content/uploads/2018/09/Absolute-Java-6th-edition.pdf>

Lecture 1.1 - Introduction to Java - Notes

Origins of the Java Language

- Well-known as a programming language for internet applications.
- Is also a general-purpose programming language suitable for most applications.
- Created by Sun Microsystems led by James Gosling in 1991.
 - Was sold to Oracle in 2010
- Originally designed for programming home appliances like washing machines and TVs.
- The Java team developed a two-step translation process to simplify the task of writing compilers for each class of appliance
 - Each appliance manufacturer uses different chips and they need something that could work on all of them
- Why was this two-step translation process so significant?
 - Writing a compiler (translation program) for each appliance would have been too expensive,
 - Instead, developed **Java byte-code, an intermediate language** that is the same for all processors.
 - Therefore, only a small, easy to write program was needed to translate byte-code into machine code for each processor
- It didn't end up being adopted by appliance manufacturers but it did become a widely used programming language.
- In 1994, Patrick Naughton and Jonothan Payne at Sun Microsystems developed a Web browser that could run Java programs over the Internet.
 - This browser evolved into one now know as HotJava
 - Started Javas connection to the Internet
- Netscape made its Web browser capable of running Java programs in 1995
 - Other companies soon followed suit

Objects and Methods

- Java is an object-oriented programming (OOP) language
 - A programming methodology that views a program as consisting of objects that interact with one another by means of action (called methods)
 - Objects of the same kind are said to have the same type or be in the same class

Terminology Comparisons

- Other high-level programming languages have constructs called procedures, methods, functions, and/or subprograms
 - All basically do the same thing which is performing actions with objects
- All programming constructs in Java are part of a class

Java Application Programs

- Two common types of Java programs are **applications** and **applets**
- A Java application program is a class with a method named main

- The main method is automatically invoked at run-time (when the program is started)
- All Java application programs start with the main method
- Can also be run with an applet viewer program for debugging
- Always use windowed interfaces
 - In contrast, application programs may use a windowing interface or console

A Simple Java Program

- A java program is a class definition with a method named main
- When the program is run the actions specified in main are carried out
- The body of the method is enclosed in curly braces {}
 - Everything that the program is going to run needs to be inside of them
- `public static void main(String[] args)`

Computer Language Levels

- High-level
 - A language that humans can read, write, and understand (words and symbols)
 - Must be translated into a language a computer can understand before it can be run
- Machine language
 - A language a computer can understand
 - Binary code
- Low-level
 - Machine language or any language similar to machine language
- Compiler
 - A program that translates a high-level language program into an equivalent low-level language program
 - Through a process called compiling

Byte-Code and the Java Virtual Machine

- The compilers for most programming languages translate high-level programs directly into the machine language for a particular computer
 - Different computers have different machine languages, therefore a different compiler is needed for each one
- In contrast, the Java compiler translates Java programs into byte-code
 - A Java program can be used on any computer, making it very portable
- Interpreter
 - The program that translates a program written in Java byte-code into the machine language for a particular computer when a Java program is executed
 - Translates and immediately executes each byte-code instruction, one after another
 - Translating byte-code into machine code is relatively easy compared to the initial compilation step

Terminology

- Code
 - Means program or part of a program
- Source code (or source program)
 - Program written in a high-level-language such as Java
 - Input to the compiler program
- Object code (or object program)
 - Translated low-level-language program
 - The output from the compiler program

Class Loader

- A program that connects the byte-code of the classes needed to run a Java program
 - In other programming languages it is sometimes called a linker
- Java programs are divided up into smaller parts called classes
 - Each class definition is normally in a separate file and is also compiled separately

Compiling a Java Program or Class

- Each class definition must be in a file whose name is the same as the class name followed by `.java`
 - The class `FirstProgram` must be in a file named `FirstProgram.java`
- Each class is compiled with the command `javac` followed by the name of the file where the class is defined
 - `javac FirstProgram.java`
- The result is a byte-code program whose filename is the same as the class name followed by `.class`
 - `FirstProgram.class`

Running a Java Program

- A Java program can be given the run command after all of its classes have been compiled
 - `java`
- Only run the main method
 - The system will automatically load and run the other classes, if there are any
- The main method in Java always begins with the line
 - `public static void main(String[] args)`
- Run the Java program by giving the command `java` followed by the name of the class containing the main method
 - `java FirstProgram`
 - No `.class` or `.java`

Syntax and Semantics

- Syntax
 - The arrangement of words and punctuations that are legal in a language
 - a.k.a the grammar rules of a language
 - For example in the english language you cannot put a full stop in the middle of a sentence or capitalise an arbitrary letter in a word
 - Describes how to write something

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